CBSE Class 10 Science Question Paper 2024

Set 1 (31/3/1) Solutions

SECTION – A

(Multiple Choice Questions)

In this section, Questions NO 1to 20 are Multiple Choice questions. All questions are Compulsory (20 x 1 = 20)

1. **Identify the product 'X' obtained in the following chemical reaction: 1**

$$CaCO_3 \xrightarrow{\Delta} X + CO_2$$

(A) Quick lime
(B) Gypsum
(C) Lime Stone
(D) Plaster of Paris

Answer: (A) Quick lime

2. **Select a pair of natural indicators from the following: 1**

(A) Litmus and methyl orange
(B) Turmeric and Litmus
(C) Phenolphthalein and methyl orange
(D) Methyl orange and Turmeric

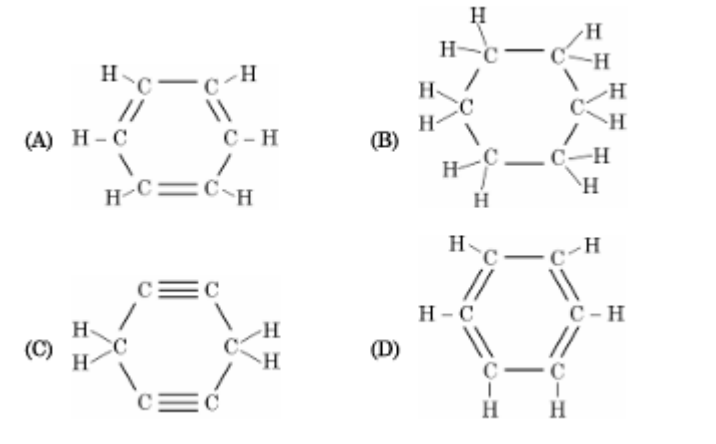
Answer: (B) Turmeric and Litmus

3. **A chemical compound used in glass, soap and paper industries is**

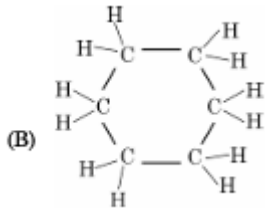
(A) Washing Soda
(B) Baking Soda
(C) Bleaching Powder
(D) Common Salt

Answer: (A) Washing Soda

4. **The structural formula of Cyclohexane is** **1**



Answer: (B)



5. **Consider the following Chemical equation:** **1**

$$aAl_2O_3 + bHCl \rightarrow cAlCl_3 + dH_2O$$

In order to balance this chemical equation, the values of a, b, c, and d must be:

(A) 1, 6, 2 and 3
(B) 1, 6, 3 and 2
(C) 2, 6, 2 and 3
(D) 2, 6, 3 and 2

Answer: (A) 1, 6, 2 and 3

6. **Which one of the following hydrocarbons is different from the others?** **1**

(A) C_4H_{10}
(B) C_7H_{14}
(C) C_5H_{12}
(D) C_2H_6

Answer: (B) C_7H_{14}

7. **Which one of the following reactions is different from the remaining three?** **1**

(A) $NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$

(B) $CaO + H_2O \rightarrow Ca(OH)_2$

(C) $KNO_3 + H_2SO_4 \rightarrow KHSO_4 + HNO_3$

(D) $ZnCl_2 + H_2S \rightarrow ZnS + 2HCl$

Answer: (B) $CaO + H_2O \rightarrow Ca(OH)_2$

8. **Select from the following a plant hormone which promotes cell division.** **1**

(A) Gibberellins
(B) Auxins
(C) Abscisic Acid
(D) Cytokinins

Answer: (D) Cytokinins

9. **Part(s) of a flower which attracts insects for pollination is (are)** **1**

(A) petals and Sepals
(B) anther and Stigma
(C) petals only
(D) sepals only

Answer: (A) petals and Sepals

10. In an experiment to study independent inheritance of two separate traits: shape and colour of seeds, the ratio of the different combinations in F_2 progeny would be **1**

(A) 1: 3
(B) 1: 2: 1
(C) 9: 3 : 3 : 1
(D) 9 : 1 : 1 : 3

Answer: (C) 9: 3: 3: 1

11. Which of the following statement(s) is (are) true about human heart? 1

(a) Right atrium receives oxygenated blood from lungs through pulmonary artery.
(b) Left atrium transfers oxygenated blood to left ventricle which sends it to various parts of the body.
(c) Right atrium receives deoxygenated blood from different parts of the body through vena cava.
(d) Left atrium transfers oxygenated blood to aorta which sends it to different parts of the body.

(A) (b) only
(B) (a) and (d)
(C) (c) and (d)
(D) (b) and (d)

Answer: (C) (c) and (d)

12. **A cross between two tall pea plants resulted in offsprings having a few dwarf plants. The gene-combination of the parental plants must be 1**

(A) Tt and Tt
(B) Tt and tt
(C) TT and tt
(D) TT and Tt

Answer: (A) Tt and Tt

13. **The phenomena of light involved in the formation of a rainbow in the sky are 1**

(A) Refraction, dispersion and reflection
(B) Refraction, dispersion and total internal reflection
(C) Dispersion, scattering and reflection
(D) Dispersion, refraction and internal reflection

Answer: (A) Refraction, dispersion and reflection

14. **In case of four wires of same material, the resistance will be minimum if the diameter and length of the wire respectively are** **1**

(A) D/2 and L/4
(B) D/4 and 4L
(C) 2D and L
(D) 4D and 2L

Answer: (D) 4D and 2L

15. A food chain will be more advantageous in terms of energy if it has **1**

(A) 2 trophic levels
(B) 3 trophic levels
(C) 4 trophic levels
(D) 5 trophic levels

Answer: (A) 2 trophic levels

16.Consider the following statements about ozone: **1**

(a) Ozone is a poisonous gas.
(b) Ozone shields the earth's surface from the infrared radiation from the sun.
(c) Ozone is a product of UV radiations acting on oxygen molecule.
(d) At the lower level of the earth's atmosphere, ozone performs most essential function.

The correct statements are:

(A) (a) and (b)
(B) (a) and (c)
(C) (b) and (c)
(D) (b) and (d)

Answer: (B) (a) and (c)

Q. Nos. 17 to 20 consists of two statements – Assertion (A) and Reason (R), answer these questions selecting the appropriate option given below:

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true and (R) is not the correct explanation of (A).
(C) (A) is true, but (R) is false.
(D) (A) is false, but (R) is true.

17. Assertion (A): A piece of Zinc metal gets reddish brown coating when kept in copper sulphate solution for some time 1

Reason (R): Copper is more reactive metal than Zinc.

Answer: (C) (A) is true, but (R) is false.

18. Assertion (A): Offsprings produced by asexual reproduction are genetically similar to the parents. 1

Reason (R): Asexual reproduction involves a single parent.

Answer: (A) Both (A) and (R) are true and (R) is the correct explanation of (A).

19. Assertion (A): Red light signals are used to stop the vehicles on the road. 1

Reason (R): Red coloured light is scattered the most so as to be visible from a large distance.

Answer: (C) (A) is true, but (R) is false.

20. Assertion (A): The waste we generate daily may be biodegradable or non-biodegradable. **1**

Reason (R): The waste generated, if not disposed off properly may cause serious environmental problems.

Answer: (A) Both (A) and (R) are true and (R) is the correct explanation of (A).

SECTION – B

Question Nos. 21 to 26 are very short answer type questions. Each question carries 2 marks.

21. When magnesium ribbon is burnt in air, an ash of white colour is produced. Write chemical equation for the reaction giving the chemical name of the ash produced. State the type of chemical reaction giving justification for your answer. **2**

Answer:
When magnesium ribbon burns in air, it reacts with oxygen to form magnesium oxide, which is the white ash.

Chemical equation:
$2Mg + O_2 \rightarrow 2MgO$

The type of chemical reaction is a combination reaction, as magnesium (Mg) combines with oxygen (O_2) to form a single product, magnesium oxide (MgO).

22. Where are auxins synthesized? How do they promote phototropism? 2

Answer:
Auxins are synthesized in the shoot tips and young leaves of plants.

They promote phototropism by accumulating on the side of the plant that is away from the light source, causing the cells on that side to elongate more than the cells on the side facing the light. This differential growth causes the plant to bend towards the light.

23. (a) **List any two pairs of visible contrasting characters of garden pea plants used by Mendel for his experiments stating the dominant and recessive characters in each pair. 2**

Answer:

1. Flower color:
   - Dominant: Purple
   - Recessive: White
2. Seed shape:
   - Dominant: Round
   - Recessive: Wrinkled

OR

(b) In human beings, the probability of getting a male or a female child is 50%. Explain with the help of a flow diagram only. 2

Answer:

- Male parent: XY

- Female parent: XX
- Possible combinations: XX (female), XY (male)

24. When do we say that a particular person is suffering from hypermetropia? List two causes of this defect. Name the type of lens used to correct this defect. 2

Answer:
A person is said to be suffering from hypermetropia (farsightedness) when they can see distant objects clearly but have difficulty seeing objects up close.

Causes:

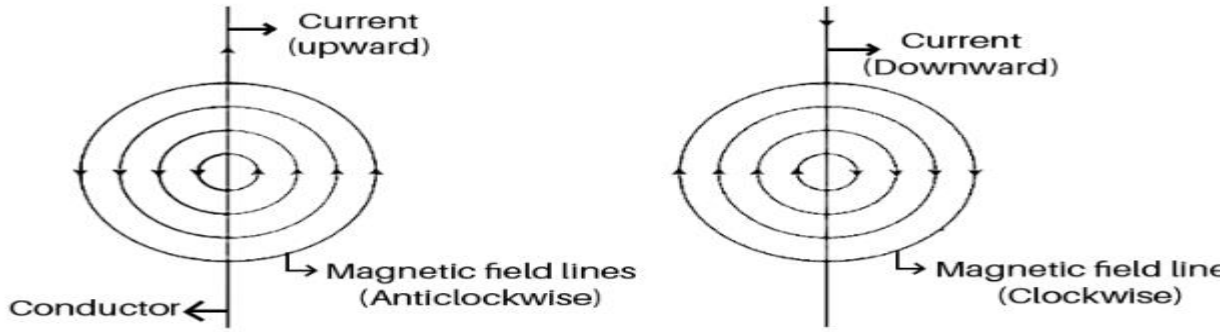
1. The eyeball is too short.
2. The lens is not curved enough.

Correction:
Convex lens

25. (a) **Draw a labelled diagram to show the pattern of magnetic field lines produced due to a current-carrying straight conductor. Mark on it the direction of current in the conductor and the direction of magnetic field lines.**

Answer:

- Direction of current: Upward
- Direction of magnetic field lines: Circular around the conductor



OR

(b) Name the device used to magnetize a piece of magnetic material. Draw a labelled diagram to show the arrangement used for the magnetisation of a cylinder made of soft iron. 2

Answer:
Device: Electromagnet

- Soft iron core
- Coils of wire
- Direction of current through coils

26. What are decomposers? List two consequences of their absence in an ecosystem. 2

Answer:
Decomposers are organisms that break down dead or decaying organisms, and in doing so, carry out the natural process of decomposition.

Consequences of their absence:

1. Accumulation of dead organic matter.
2. Disruption of nutrient cycling, leading to soil infertility.

Section C

Question No 27 to 33 are short answer type questions each questions carries 3 marks

27. **State reasons for the following:** **3**

(a) Zinc oxide is an amphoteric oxide.
Answer: Zinc oxide reacts with both acids and bases to form salts and water, indicating it has both acidic and basic properties, making it amphoteric.

(b) Sodium metal is stored in a bottle filled with kerosene oil.
Answer: Sodium is highly reactive with water and oxygen, which can cause it to ignite. Storing it in kerosene prevents it from coming into contact with moisture and air, thus preventing accidental ignition.

(c) In the reactions of nitric acid with metals, generally hydrogen gas is not evolved.
Answer: Nitric acid is a strong oxidizing agent, which means that instead of hydrogen gas, nitrogen oxides are typically produced during the reaction with metals.

28. (a) **State giving reason the reduction process to obtain the following metals from their compounds:** **3**

(i) Mercury
Answer: Mercury can be obtained by heating its sulfide ore (cinnabar) in air, which decomposes to form mercury and sulfur dioxide.

(ii) Copper
Answer: Copper can be extracted by heating its sulfide ore (copper(I) sulfide) in the presence of oxygen, producing copper and sulfur dioxide.

(iii) Sodium
Answer: Sodium is obtained by the electrolysis of molten sodium chloride (Downs process), which separates sodium metal and chlorine gas.

OR

28. (b) State giving reason for the change in appearance observed when each of the following metal is exposed to atmospheric air for some time: 3

(i) Silver
Answer: Silver tarnishes, forming a black coating of silver sulfide due to reaction with sulfur compounds in the air.

(ii) Copper
Answer: Copper develops a green patina (copper carbonate) due to prolonged exposure to carbon dioxide and moisture in the air.

(iii) Iron
Answer: Iron rusts, forming a reddish-brown coating of iron oxide due to reaction with oxygen and moisture in the air.

29. We water the soil, but it reaches the topmost leaves of the plants. Explain in brief the process involved. 3

Answer:
The process involved is transpiration. Water is absorbed by the roots from the soil and moves upwards through the plant via the xylem vessels. This movement is driven by capillary action and root pressure. When water reaches the leaves, it evaporates into the air through tiny openings called stomata, creating a negative pressure that pulls more water up from the roots. This continuous movement ensures that water reaches even the topmost leaves.

30. (a) **List two constituents of Central Nervous System (CNS). How are these components protected from injuries? 3**

Answer:

- Constituents: Brain and Spinal Cord
- Protection:
  - The brain is protected by the skull (cranium) and surrounded by cerebrospinal fluid (CSF) which acts as a cushion.
  - The spinal cord is protected by the vertebral column (spine) and also surrounded by CSF.

(b) Write two limitations of the use of electrical impulses. 3

Answer:

- Electrical impulses cannot be transmitted over long distances without loss of signal.
- Electrical impulses can only travel at a fixed speed and cannot be modulated for different rates of response.

31. Name and explain the phenomenon of light due to which the path of a beam of light becomes visible when it enters a smoke-filled room through a small hole. Also state the dependence of the colour of the light we receive on the size of the particle of the medium through which the beam of light passes. 3

Answer:

- Phenomenon: Tyndall effect (scattering of light)

- Explanation: The path of the beam of light becomes visible because the particles in the smoke scatter the light, making the beam visible.
- Dependence on particle size:
  - Smaller particles scatter shorter wavelengths of light (blue/violet) more effectively.
  - Larger particles scatter longer wavelengths of light (red/orange) more effectively.

32. Explain in brief the function of an electric fuse in a domestic circuit. An electric heater of current rating 3 kW; 220 V is to be operated in an electric circuit of rating 5 A. What is likely to happen when the heater is switched 'ON'? Justify your answer with necessary calculation. 3

Answer:

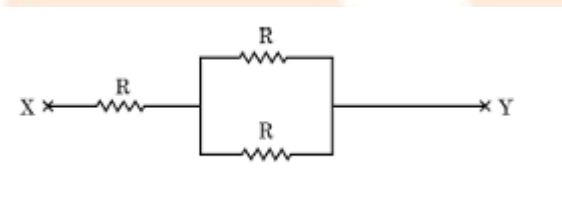
- Function of fuse: An electric fuse protects the circuit by breaking the circuit if the current exceeds a safe level, preventing overheating and potential fires.
- Calculation:
  - Power (P) = 3 kW = 3000 W
  - Voltage (V) = 220 V
  - Current (I) = P / V = 3000 W / 220 V ≈ 13.64 A
- Justification: When the heater is switched on, it will draw a current of approximately 13.64 A, which exceeds the 5 A rating of the circuit. This will cause the fuse to blow, breaking the circuit to prevent damage.

33. (a) **State Ohm's law. Write formula for the equivalent resistance R_p of the parallel combination of three resistors of values R_1, R_2, and R_3.**

Answer:

- Ohm's Law: The current through a conductor between two points is directly proportional to the voltage across the two points, provided the temperature remains constant. V=IR
- Formula for equivalent resistance in parallel: $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$

(b) **Find the resistance of the following network of resistors:**



Answer:

- Assuming all resistors have the same value R:
- The two resistors in series (top branch) have an equivalent resistance of 2R.
- This equivalent resistance is in parallel with the single resistor RRR in the bottom branch.
- The equivalent resistance R_p of the parallel combination is:

$$\frac{1}{R_p} = \frac{1}{2R} + \frac{1}{R} = \frac{1}{2R} + \frac{2}{2R} = \frac{3}{2R}$$

$$R_p = \frac{2R}{3}$$

SECTION – D

Question Nos. 34 to 36 are long answer type questions. Each question carries 5 marks.

34. (a) (i) Five solutions A, B, C, D and E when tested with pH paper showed pH as 4, 1, 8, 7 and 10 respectively. Which solution is: 5

(1) Strongly acidic
(2) Strongly alkaline
(3) Weakly acidic
(4) Neutral
(5) Weakly alkaline? Arrange the solutions in increasing order of H^+ ion concentration.

Answer:

- Strongly acidic: B (pH 1)
- Strongly alkaline: E (pH 10)
- Weakly acidic: A (pH 4)
- Neutral: D (pH 7)
- Weakly alkaline: C (pH 8)

Increasing order of H^+ ion concentration: E (pH 10) < C (pH 8) < D (pH 7) < A (pH 4) < B (pH 1)

(ii) **Write the name and formula of (1) an acidic salt and (2) a basic salt giving the name of the parent acid and parent base used to form the salt in each case.**

Answer:

1. Acidic salt: Sodium hydrogen sulfate ($NaHSO_4$)
    - Parent acid: Sulfuric acid (H_2SO_4)
    - Parent base: Sodium hydroxide ($NaOH$)
2. Basic salt: Sodium carbonate (Na_2CO_3)
    - Parent acid: Carbonic acid (H_2CO_3)
    - Parent base: Sodium hydroxide ($NaOH$)

OR

34. (b) **Name and state in brief the process which is used to prepare sodium hydroxide from sodium chloride. In this process along with the main product two gases 'X' and 'Y' are also given off at the two electrodes. Name 'X' and 'Y' specifying the name of their respective electrode at which each gas is obtained. One of these gases when reacts with dry calcium hydroxide produces a compound 'Z' which is widely used in water treatment plants and textile industries. Name 'Z' and write chemical equation for the reaction involved in its formation.**

5

Answer:

- Process: Electrolysis of brine (Chloralkali process)
- Gas X: Hydrogen (H_2) - obtained at the cathode
- Gas Y: Chlorine (Cl_2) - obtained at the anode

- Compound Z: Calcium hypochlorite ($Ca(OCl)_2$)

- Reaction for formation of Z:
  $Cl_2 + Ca(OH)_2 \rightarrow Ca(OCl)_2 + H_2O$

35. (a) (i) **What are spores? On which structures are they formed? How do they overcome unfavourable conditions? Name the organism which multiplies with the help of these structures. 5**

Answer:

- Spores: Spores are asexual reproductive units capable of developing into a new organism without fusion with another cell.

- Formed on: Spores are typically formed on structures called sporangia.

- Overcome unfavourable conditions: Spores can withstand extreme conditions such as heat and dryness due to their hard protective coat.

- Organism: Fungi (e.g., Rhizopus) and some bacteria.

(ii) **Give two reasons why some plants are grown by the method of vegetative propagation. List two methods used to grow plants vegetatively.**

Answer:

- Reasons for vegetative propagation:

  1. It ensures the production of plants that are genetically identical to the parent plant.

  2. It allows plants to reproduce without the need for seeds, which may be sterile or difficult to germinate.

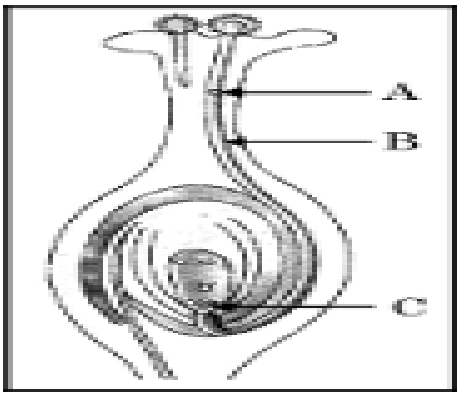
- Methods of vegetative propagation:

    1. Cutting: A piece of the stem, root, or leaf is cut from the parent plant and planted to grow a new plant.

    2. Grafting: A part of one plant (scion) is attached to another plant (rootstock) so that they grow together.

OR

35. (b) (i) Study the diagram given below and name the parts marked as A, B and C. What happens when B reaches C in the ovary? Mention its significance. 5



Answer:

- **A:** Stigma
- **B:** Pollen tube
- **C:** Ovule

What happens when B (pollen tube) reaches C (ovule) in the ovary:

When the pollen tube (B) reaches the ovule (C), it releases the male gametes (sperm cells). One of the male gametes fuses with the egg cell in the ovule, resulting in fertilization. This leads to the formation of a zygote, which will eventually develop into an embryo.

Significance:

- The fusion of male and female gametes initiates the process of seed formation.
- The fertilized ovule develops into a seed, which can grow into a new plant.
- The surrounding ovary develops into a fruit, which protects the seed and helps in its dispersal.

(ii) Write the post-fertilisation changes that occur in a flower.

Answer:

- **Formation of Seed:** The fertilized ovule develops into a seed. The zygote forms an embryo, and the ovule's integuments form the seed coat.
- **Formation of Fruit:** The ovary develops into a fruit, which encloses the seeds. The fruit aids in protecting the seeds and in their dispersal.
- **Degeneration of Floral Parts:** The petals, sepals, stamens, and style usually wither and fall off as they are no longer needed.
- **Endosperm Development:** The endosperm forms, providing nourishment to the developing embryo.
- **Seed Maturation:** The seed undergoes various changes to become mature and ready for germination. This includes dehydration and hardening of the seed coat.

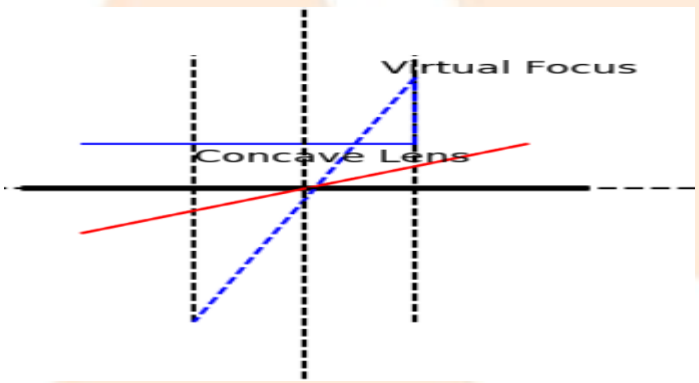
36. (a) (i) Draw a ray diagram to show the path of the refracted ray in each of the following cases: 5

A ray of light incident on a concave lens

(1) parallel to its principal axis, and

(2) is directed towards its principal focus.

Answer:



(ii) A 4 cm tall object is placed perpendicular to the principal axis of convex lens of focal length 24 cm. The distance of object from the lens is 16 cm. Find the position and size of image formed.

(a) (ii) Image Formation by Convex Lens

Given:

- Object height (h_o) = 4 cm
- Focal length (f) = 24 cm
- Object distance (u) = 16 cm

To Find:

- Image distance (v)
- Image height (h_i)

Solution:

1. **Use the Lens Formula:**

$$\frac{1}{f}=\frac{1}{v}-\frac{1}{u}$$

Substituting the given values:

$$\frac{1}{24}=\frac{1}{v}-\frac{1}{16}$$

Solve for $\frac{1}{v}$:

$$\frac{1}{v}=\frac{1}{24}+\frac{1}{16}$$

Finding a common denominator:

$$\frac{1}{v}=\frac{2}{48}+\frac{3}{48}=\frac{5}{48}$$

v1=482+483=485

$$v=\frac{48}{5}=9.6\text{ cm}$$

2. **Calculate the Image Height (h_i):** Use the magnification formula:

$$\text{Magnification}(M)=\frac{h_i}{h_o}=\frac{v}{u}$$

Substituting the known values:

$$M = \frac{9.6}{16} = 0.6M$$
$$M \times h_o = 0.6 \times 4 = 2.4 \text{ cm}$$

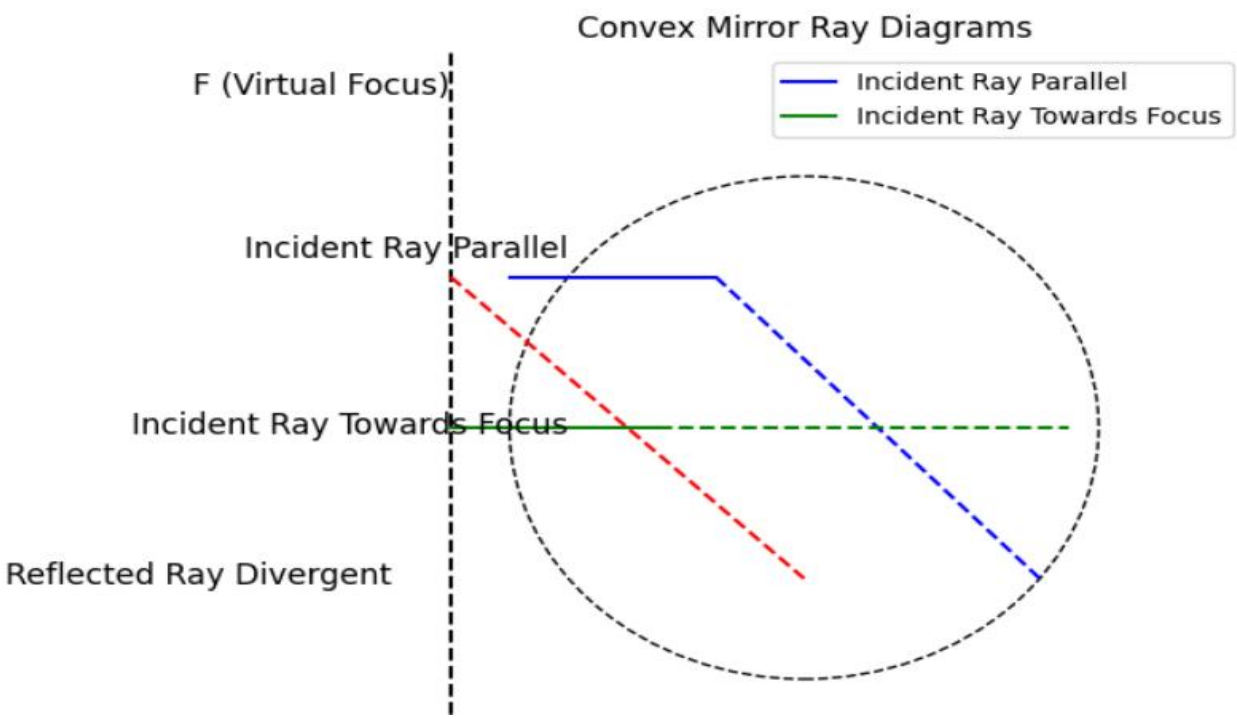
OR

36. (b) (i) Draw a ray diagram to show the path of the reflected ray in each of the following cases: 5

A ray of light incident on a convex mirror

(1) parallel to its principal axis, and

(2) is directed towards its principal focus



(ii) 1.5 cm tall candle flame is placed perpendicular to the principal axis of a concave mirror of focal length 12 cm. If the distance of the flame from the pole of the mirror is 18 cm, use mirror formula to determine the position and size of the image formed.

Given:

- Object height (h_o) = 1.5 cm
- Focal length (f) = -12 cm (negative for concave mirror)
- Object distance (u) = -18 cm (negative as per sign convention)

To Find:

- Image distance (v)
- Image height (h_i)

Mirror Formula:

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

Magnification Formula:

$$m = \frac{h_i}{h_o} = -\frac{v}{u}$$

Calculations:

1. **Using the Mirror Formula to Find v:**

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

Substituting the given values:

$$\frac{1}{-12} = \frac{1}{v} + \frac{1}{-18}$$

$$\frac{1}{v} = \frac{1}{-12} + \frac{1}{18}$$

Finding a common denominator:

$$\frac{1}{v} = \frac{-3+2}{36} = \frac{-1}{36}$$

$$v = -36\ cm$$

2. **Using the Magnification Formula to Find h_i:**

$$m = -\frac{v}{u}$$

$$m = -\frac{-36}{-18} = 2m$$

$$h_i = m \times h_o = 2 \times 1.5 = 3 \text{ cm}$$

SECTION – E

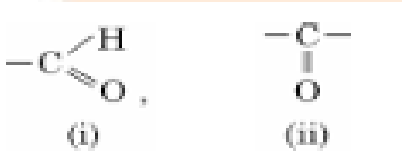
Question Nos. 37 to 39 are Case/Source based questions. Each question carries 4 marks.

37. More than three million carbon compounds have been discovered in the field of chemistry. The diversity of these compounds is due to the capacity of carbon atoms for bonding with one another as well as with other atoms. Most of the carbon compounds are poor conductors of electricity and have low melting and boiling points. 4

(a) Write the molecular formula of the first two members of homologous series having functional group –Br.

1. CH_3Br (Bromomethane)
2. CH_3CH_2Br (Bromoethane)

(b) Given below are the formulae of some functional groups:



- (i) –CHO
- (ii) –CO

Write the name of these functional groups.

1. (i) Aldehyde
2. (ii) Ketone

(c) What would be observed on adding a 5% alkaline potassium permanganate drop by drop to some warm ethanol taken in a test tube? State the role of $KMnO_4$ in the reaction and write the chemical equation for the reaction involved.

- **Observation:** The purple colour of $KMnO_4$ will gradually fade.
- **Role of $KMnO_4$:** $KMnO_4$ acts as an oxidizing agent.
- **Chemical Equation:** $3C_2H_5OH + 2KMnO_4 + 4KOH \rightarrow 3CH_3COOH + 2MnO_2 + 2H_2$

OR

Write the name of the compound formed when ethanol is heated at 443 K temperature with excess of conc. $H2SO_4$. What is the role of conc. $H2SO_4$ in the reaction? Write the chemical equation for the reaction involved.

- **Name of the Compound Formed:** Ethene ($C2H_4$)
- **Role of conc. H_2SO_4:** Concentrated H_2SO_4 acts as a dehydrating agent.
- **Chemical Equation:** $C_2H_5OH \xrightarrow[\Delta]{Conc.H_2SO_4} C_2H_4 + H_2O$

38. Human digestive system is a tube running from mouth to anus. Its main function is to breakdown complex molecules present in the food which cannot be absorbed as such into smaller molecules. These molecules are absorbed across the walls of the tube and the absorbed food reaches each and every cell of the body where it is utilised for obtaining energy. 4

(a) Name the glands present in the buccal cavity and write the components of food on which the secretion of these glands act upon.

- **Glands:** Salivary glands
- **Components:** Carbohydrates

(b) Two organs have a sphincter muscle at their exit. Name them.

1. **Stomach (Pyloric sphincter)**
2. **Anus (Anal sphincter)**

(c) What will happen if:

- (i) Mucus is not secreted by the gastric glands.

  Answer: The stomach lining will be damaged by the acidic environment, leading to ulcers.

- (ii) Villi are absent in the small intestine.

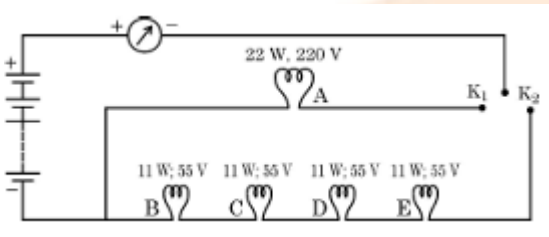
  Answer: Absorption of nutrients will be severely reduced, leading to malnutrition.

OR

(c) "Bile juice does not contain any enzyme, yet it has important roles in digestion." Justify the statement.

- **Answer:** Bile juice emulsifies fats, breaking them into smaller droplets, which increases the surface area for the action of lipase. It also neutralizes the acidic chyme from the stomach, creating an optimal pH for enzymatic activity in the small intestine.

39. In a domestic circuit five LED bulbs are arranged as shown. The source voltage is 220 V and the power rating of each bulb is marked in the circuit diagram. Based on the following circuit diagram, answer the following questions: 4



(a) State what happens when:

- (i) Key K_1 is closed.

  Answer: When K_1 is closed, bulbs A and B will light up as they are in parallel and receive the full voltage of 220 V.

- (ii) Key K_2 is closed.

  Answer: When K_2 is closed, bulbs C, D, and E will light up. These bulbs are in parallel with each other and in series with the combination of A and B. However, since the keys K_1 and K_2 control different parts of the circuit, K_2 affects only the bulbs in its own parallel connection.

(b) Find the current drawn by the bulb B when it glows.

Given:

- Power rating of bulb B (P) = 55 W
- Source voltage (V) = 220 V

$$I = \frac{P}{V}$$

$$I = \frac{55\ \text{W}}{220\ \text{V}}$$

$$\text{A}I = 0.25\ A$$

Answer: The current drawn by bulb B when it glows is 0.25 A.

(c) Calculate:

(i) The resistance of bulb B

Using Ohm's Law and the power formula: $P = \frac{V^2}{R}$

$$R = \frac{V^2}{P}$$

$$R = \frac{(220\ \text{V})^2}{55\ \text{W}}$$

$$R = \frac{48400}{55}$$

$$\text{O}R = 880\ O$$

Answer: The resistance of bulb B is 880 Ω.

(ii) Total resistance of the combination of four bulbs B, C, D, and E.

Given:

- All bulbs have the same power rating and thus the same resistance.

- The resistance of each bulb (R_B, R_C, R_D, R_E) = 880 Ω (as calculated for bulb B)

Bulbs C, D, and E are in parallel:

$$\frac{1}{R_{\text{parallel}}} = \frac{1}{R_C} + \frac{1}{R_D} + \frac{1}{R_E}$$

$$\frac{1}{R_{\text{parallel}}} = \frac{1}{880} + \frac{1}{880} + \frac{1}{880}$$

$$\frac{1}{R_{\text{parallel}}} = \frac{3}{880}$$

$$R_{\text{parallel}} = \frac{880}{3}$$

$$R_{\text{parallel}} \approx 293.33 \text{ O}$$

This parallel combination is in series with bulb B:

$$R_{\text{total}} = R_B + R_{\text{parallel}}$$

$$R_{\text{total}} = 880 \text{ O} + 293.33 \text{ O}$$

$$R_{\text{total}} \approx 1173.33 \text{ O}$$

Answer: The total resistance of the combination of four bulbs B, C, D, and E is approximately 1173.33 Ω.

OR

(c) What would happen to the glow of all the bulbs in the circuit when keys K_1 and K_2 both are closed and the bulb C suddenly gets fused? Give reason to justify your answer.

Answer:

- When both K_1 and K_2 are closed, all the bulbs A, B, C, D, and E are intended to light up.
- If bulb C gets fused, it effectively breaks the parallel circuit comprising bulbs C, D, and E.
- Bulbs D and E will also go out because the parallel path is broken, and there will be no current through these bulbs.
- Bulbs A and B, however, will remain unaffected and will continue to glow as they are on a different parallel path controlled by K_1.

Reason:

- The bulbs C, D, and E are in parallel with each other and in series with the combined parallel circuit of A and B.
- Fusing bulb C breaks the circuit for D and E, but does not affect the path for A and B as they are independent of the C, D, E parallel combination.